Experiment 1: Non-Visual Baseline and Estimator Comparison

Raw write-up for Chapter 4, Section 4.1

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Reporting cutoff: 6 November 2026

Purpose of this document. This is the raw, complete write-up of Experiment 1, written before the thesis chapter; Section 4.1 of the final report is extracted from it. It carries both result slots, every table and figure, and the analysis. Every number traces to a file under `output/` named in the slot's "Reporting rules" paragraph. Open concerns are set in red where they arise.

4.1.1 Simulation

Status: completed. Four `dataset_allsensor_realistic` recordings, two layouts each recorded without and with warehouse traffic; wheel odometry, IMU dead reckoning, the EKF at the visual systems' own inertial noise configuration, a measured-noise EKF as diagnostic, VINS-Fusion, and ORB-SLAM3 with the GPU front end in two loop-closer states, twenty-eight runs, all over the full common interval. Re-scored on 7 November 2026 after the earlier `allsensor` result was retired with its datasets and one misfiled ORB-SLAM3 run was set aside and re-run.

Objective. Establish whether a non-visual estimator provides usable localisation on the same recordings the visual pipelines consumed, and whether each estimator's reported uncertainty describes its error.

Process. `script/proprio_estimator.py calibrate` measured each recording's wheel and inertial biases and wheel noise against ground truth from its stationary head and its driven length; `run` produced the wheel, IMU and EKF trajectories offline; `script/analyse_proprio.py` scored them with the live VINS-Fusion and ORB-SLAM3 runs over the common interval. The EKF reported here does *not* use the measured inertial noise: by the design decision of 6 November 2026 its three inertial noise terms are the values both visual configurations declare (`config/wil_sim/stereo_imu.yaml`: gyroscope density 0.002, accelerometer density 0.02, gyroscope bias walk 1×10^{-4}), converted to per-sample σ at 200 Hz by `script/make_vslam_noise_yaml.py`, so that the proprioceptive and visual families run on the same hand-set inertial model; biases and both wheel σ have no visual counterpart and stay measured. The measured-noise filter is run beside it as a diagnostic row and is not the baseline.

Datasets. Table 1.1: the two static twins and the two traffic recordings of the realistic worlds, both layouts. Traffic is noted as a condition of a recording, not treated as a factor, since no two recordings share a route.

Reporting rules. Aligned ATE is RMSE after a rigid $SE(3)$ fit with scale fixed, the convention of every slot; anchored error is the direct difference from ground truth with no alignment and is defined only for the estimators that start from the ground-truth pose; aligned rotation is the geodesic angle after the same fit, defined for every estimator; ANEES is the average normalised estimation error squared over decorrelated

Table 1.1: Recordings of Experiment 1 (names without the `dataset_allsensor_` prefix), from the bags. Route is the ground-truth path length; samples are ground-truth odometry messages.

Recording	Layout	Traffic	Route [m]	Span [s]	Samples
`realistic_000`	U	none (static twin)	171.3	125.5	6 278
`realistic_001`	I	none (static twin)	133.0	96.9	4 847
`realistic_002`	U	pallet jacks and people	102.7	79.5	3 975
`realistic_003`	I	pallet jacks and people	158.0	117.0	5 852

1 Hz samples, 3.0 for a consistent planar filter. Ground truth is the bag's `/ground_truth/odometry` moved from the model origin to the IMU point every estimator reports (lever arm 0.338 m forward, 0.192 m up). The visual runs are pinned by name: VINS-Fusion `cpu`; ORB-SLAM3 `loopclosing_gpu` and `noloopclosing_gpu`, both with the GPU feature extractor, at replay rate 1.0. The loop-closer state is read from each run's `loop_closing.csv` (patch-7 signature: sub-millisecond rows mean a disabled closer), not from the run name. On `realistic_002` the directory named `loopclosing_gpu` first held a run whose manifest names the `realistic_003` bag (3 535 stereo pairs received against the 2 409 that recording holds); it was scored against the wrong truth for a day, is set aside under `trash/output/misfiled_20261106/`, and the run reported here was made on the right bag on the evening of 6 November (2 382 pairs received). Sources: `output/compare/simulation/exp1-realistic/` (`proprio_metrics.csv`, `proprio_rpe_metrics.csv`, the noise models under `noise/`), run directories under `output/output_{wheel,imu,ekf,ekf_measured}/simulation/`. One run per cell, $n = 1$.

Results. At the visual systems' inertial noise the EKF holds 0.68–2.80 m aligned ATE and is behind VINS-Fusion on all four recordings; the measured-noise diagnostic holds 0.21–0.58 m. Table 1.2 gives every estimator on every recording, table 1.3 the relative pose error at 1 s, and figure 1.1 the trajectories.

Analysis. With both families on the same hand-set inertial model the ordering is settled: VINS-Fusion is the most accurate estimator on every recording (0.22–0.66 m) and the EKF, at 0.68–2.80 m, is behind it on all four and behind the better ORB-SLAM3 run on three. The measured-noise diagnostic puts the same filter at 0.21–0.58 m, ahead of VINS-Fusion on three of four, so the whole of the EKF's deficit is the noise model and none of it the architecture — the September control, reproduced on new recordings and larger.

The mechanism is the one isolated in September. The configuration's gyroscope density is five to seven times the measured value, which inflates the heading process noise forty-fold; with the heading unobserved except through the wheel yaw-rate residual, that residual steers the heading instead of the bias, and yaw RMS rises from 0.4–5.7° to 4.5–28.0°. On routes of 100 to 170 m with repeated turns that heading error is what the aligned ATE measures.

ORB-SLAM3's divergences are not a loop-closure effect. With the GPU front end six of eight runs hold 0.17–0.52 m and two diverge, to 4.78 m with the closer active on `realistic_001` and to 1.91 m with it disabled on `realistic_003`, while the other closer state on each of those recordings holds 0.40 and 0.50 m. Each divergent run carries three or seven jumps (table 1.3) and its clean relative error is as low as the stable runs', so the error is a few discrete events, not steady-state tracking. Which front-end condition produces them is not isolated here; the CPU runs of the same recordings belong to Experiment 3, where the same pattern recurs.

Traffic appears as a condition, not a tested factor. VINS-Fusion is 0.66 m on the U traffic recording against 0.29 m on its static twin and 0.27 m against 0.22 m on the I layout; the proprioceptive estimators are unaffected by construction. The U recordings do not share a route, so the difference cannot be credited to traffic alone.

The complementary failure of the two proprioceptive sensors holds. The IMU keeps heading to

Table 1.2: Estimator comparison on the four realistic recordings. The closer state of each ORB-SLAM3 run is read from its artefact. ANEES is reported for the proprioceptive estimators, which carry a covariance; a consistent filter gives 3.0, and every value here is above it. **Bold** marks the lowest aligned ATE of a recording; the measured-noise filter is a diagnostic row, not the baseline, so it is shown bold where it is lowest but is not the slot's result.

Recording	Estimator	Closer	Anch. RMS [m]	Anch. final [m]	Aligned ATE [m]	Aligned rot. [°]	Yaw RMS [°]	ANEES
`realistic_000`	Wheel odometry	–	5.89	5.66	2.56	9.14	18.05	1810
	IMU dead reckoning	–	36.19	76.01	23.02	177.21	0.84	454
	EKF, VSLAM-config noise	–	2.49	3.42	1.34	5.83	10.62	19
	EKF, measured noise (diag.)	–	0.33	0.35	**0.23**	0.47	0.43	111
	VINS-Fusion	–	–	–	0.29	1.85	–	–
	ORB-SLAM3 GPU, LC run	active	–	–	0.29	12.80	–	–
	ORB-SLAM3 GPU, no-LC run	disabled	–	–	0.52	12.84	–	–
`realistic_001`	Wheel odometry	–	1.40	2.90	1.03	6.04	7.27	176
	IMU dead reckoning	–	15.40	39.82	13.05	0.25	0.55	160
	EKF, VSLAM-config noise	–	0.94	0.83	0.68	3.81	4.51	8
	EKF, measured noise (diag.)	–	0.26	0.26	**0.21**	0.33	0.36	31
	VINS-Fusion	–	–	–	0.22	1.18	–	–
	ORB-SLAM3 GPU, LC run	active	–	–	4.78	18.98	–	–
	ORB-SLAM3 GPU, no-LC run	disabled	–	–	0.40	15.13	–	–
`realistic_002`	Wheel odometry	–	6.54	6.08	4.81	17.03	25.00	1778
	IMU dead reckoning	–	25.88	39.17	19.01	118.74	1.91	695
	EKF, VSLAM-config noise	–	2.06	3.29	1.58	9.92	13.65	37
	EKF, measured noise (diag.)	–	0.59	0.30	0.37	0.70	1.03	191
	VINS-Fusion	–	–	–	0.66	3.57	–	–
	ORB-SLAM3 GPU, LC run	active	–	–	0.40	16.04	–	–
	ORB-SLAM3 GPU, no-LC run	disabled	–	–	**0.17**	16.08	–	–
`realistic_003`	Wheel odometry	–	6.71	7.19	4.30	19.24	35.82	3399
	IMU dead reckoning	–	13.72	35.47	11.60	180.00	1.68	140
	EKF, VSLAM-config noise	–	4.44	6.23	2.80	15.92	28.01	158
	EKF, measured noise (diag.)	–	0.79	1.49	0.58	3.90	5.72	106
	VINS-Fusion	–	–	–	**0.27**	1.51	–	–
	ORB-SLAM3 GPU, LC run	active	–	–	0.50	13.16	–	–
	ORB-SLAM3 GPU, no-LC run	disabled	–	–	1.91	13.53	–	–

Table 1.3: Relative pose error at $\Delta t = 1\,\mathrm{s}$, RMSE over all pairs and over pairs that span no detected jump (a step above $10\,\mathrm{m\,s^{-1}}$ between consecutive poses); the jump count is per run. **Bold** marks the lowest jump-free translational RPE of a recording.

Recording	Estimator	Jumps	Translation [m]		Rotation [°]	
			all pairs	clean	all pairs	clean
`realistic_000`	Wheel odometry	0	0.028	0.028	1.720	1.720
	IMU dead reckoning	0	1.941	1.941	0.023	0.023
	EKF	44	0.240	0.218	0.979	0.893
	EKF	0	0.039	0.039	0.098	0.098
	VINS-Fusion	0	0.021	0.021	0.443	0.443
	ORB-SLAM3 GPU, LC run	3	0.246	0.017	5.902	0.094
	ORB-SLAM3 GPU, no-LC run	3	0.254	**0.013**	5.909	0.090
`realistic_001`	Wheel odometry	0	0.033	0.033	1.733	1.733
	IMU dead reckoning	0	0.980	0.980	0.023	0.023
	EKF	42	0.254	0.202	0.964	0.811
	EKF	0	0.045	0.045	0.090	0.090
	VINS-Fusion	0	0.018	**0.018**	0.366	0.366
	ORB-SLAM3 GPU, LC run	7	1.191	0.034	7.031	0.120
	ORB-SLAM3 GPU, no-LC run	2	0.302	0.020	6.845	0.155
`realistic_002`	Wheel odometry	0	0.027	0.027	1.997	1.997
	IMU dead reckoning	0	1.505	1.505	0.033	0.033
	EKF	67	0.243	0.158	0.985	0.780
	EKF	0	0.035	0.035	0.075	0.075
	VINS-Fusion	0	0.038	0.038	0.790	0.790
	ORB-SLAM3 GPU, LC run	4	0.463	**0.017**	10.403	0.084
	ORB-SLAM3 GPU, no-LC run	1	0.147	0.028	7.422	0.070
`realistic_003`	Wheel odometry	0	0.023	0.023	1.447	1.447
	IMU dead reckoning	0	0.936	0.936	0.023	0.023
	EKF	22	0.202	0.185	1.026	0.969
	EKF	0	0.036	0.036	0.154	0.154
	VINS-Fusion	0	0.014	**0.014**	0.327	0.327
	ORB-SLAM3 GPU, LC run	3	0.165	0.028	6.068	0.174
	ORB-SLAM3 GPU, no-LC run	3	0.151	0.036	6.088	0.333

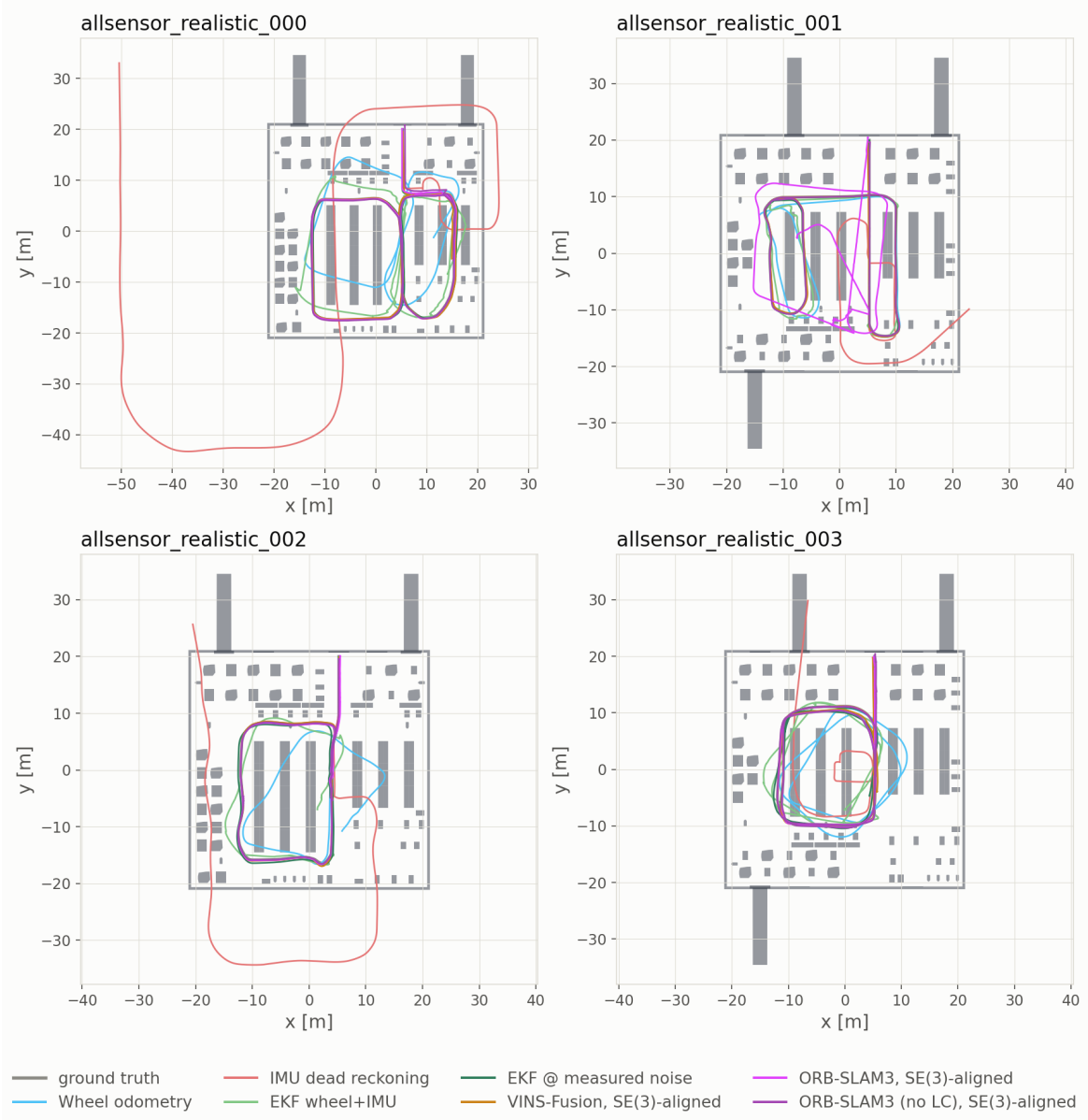


Figure 1.1: Estimator comparison on every realistic recording, world frame: the dead-reckoning estimators as estimated from the ground-truth start pose, the visual estimators after the rigid $SE(3)$ alignment used for their metrics, one legend for all four panels. The IMU leaves the building on every recording; the EKF at the visual systems' noise (light green) wanders in heading where the measured-noise filter (dark green) stays on the route; ORB-SLAM3's divergent runs are the excursions on `001` (loop closer active) and `003` (closer disabled).

0.5–1.9° RMS and loses position by 12–23 m; the wheels keep position to 1.0–4.8 m and lose 7–36° of heading. Every proprioceptive covariance is overconfident: ANEES 8–3399 against 3.0, the EKF at the configuration noise least so because its inflated process noise is closer to the error it actually makes.

Limitations, each stated once. *n:* one run per cell, no uncertainty; ORB-SLAM3's two divergences out of eight are the only run-to-run figure in the slot. *Equal handicap, not equal effort:* the shared inertial model was fitted to neither family. *Idealised wheels:* simulated encoders with slip under 1 %, the proprioceptive estimators' best case. *Route and traffic confounded:* no two recordings share a route. *Run bookkeeping:* one directory held a run on another recording for a day; the manifest caught it, and the launch should refuse an output directory whose name disagrees with its dataset.

[Resolved: `realistic_002/loopclosing_gpu`] Earlier on 6 November this run was reported as closure-free from its `loop_closing.csv` and queried. Its manifest shows the real cause: it was launched on the `realistic_003` bag with `realistic_002`'s output path, so it was scored against the wrong truth (4.94 m, 48.9°, 139 % of the recording's frames processed). It is set aside, not deleted; the configuration was re-run on the right bag the same evening and every table here carries that run.

Summary. On the realistic recordings IMU-only dead reckoning drifts by 12–23 m and wheel odometry by 1.0–4.8 m of aligned ATE; the EKF, run on the visual systems' own inertial noise as the design requires, holds 0.68–2.80 m and loses to VINS-Fusion on every recording, while the same filter on a measured noise model would hold 0.21–0.58 m. VINS-Fusion holds 0.22–0.66 m; ORB-SLAM3 with the GPU front end holds 0.17–0.52 m on six runs of eight and diverges on two, one in each closer state. Against the project's target of 0.05 m ATE no estimator qualifies; against the 1.5 m RPE bound at 1 s, all but the IMU do.

Future plan. Fit the visual systems' inertial noise to this platform rather than copying a file, since the slot shows the noise model is worth more than the estimator; feed the VIO pose into the EKF as a third measurement; and bridge the real robot's wheel speed in metric units so that this slot's real-robot half can be formed.

4.1.2 Real robot

Status: partial. Four real recordings with an independent planar reference (`dataset_realv6_000`, `dataset_realv7_000` to `_002`). A non-visual baseline now exists: wheel+gyro dead reckoning with the feedback speed scaled by a factor fitted once against the tracker on `realv7_000` and held on the other three. On every held recording it is at least as accurate as the best ORB-SLAM3 run of Experiment 3 (0.22–0.46 m against 0.44–0.52 m planar ATE). The EKF of the simulation slot cannot be formed, because the rig publishes no steering measurement; every value is provisional until the marker-to-IMU lever arm is measured.

Objective. The real-robot half of the question of section 4.1.1: does a non-visual estimator provide usable localisation on the recordings the visual pipelines consumed, and do the visual pipelines earn their place over it.

Process. The bags carry `/feedback/velocity` and `/feedback/steering` (50 Hz, `std_msgs/Float64`) and the WitMotion yaw sensor on `/hwt101ct_yaw_publisher` (200 Hz). The velocity is in a firmware unit, not metres per second; the steering is a command echo that takes two values (0 and −30) on every recording and so carries no yaw rate; the yaw sensor's orientation field is identity and its rate on the y axis is the live gyro (the sensor is mounted with y vertical: that axis integrates to 6.29 rad over `realv7_000`, whose tracker heading sweeps 6.3 rad; x and z integrate to under 0.05 rad).

`script/real_wheel_baseline.py` integrates heading from that gyro with its rest bias removed (3–7 s at rest at the head of each bag) and position from the scaled speed. The scale k, the gyro axis and its sign are fitted on `realv7_000` alone by minimising the planar ATE after rigid $SE(2)$ alignment with scale fixed (a wrong sign mirrors the path, which a proper rotation cannot align), then held: the fit recording's own score is a calibration, the three held scores are the result. The fitted $k = 0.325$ agrees with the plain path-length ratio of each `realv7` bag (0.323 to 0.329). The time offset between tracker and bag is the wheel-speed cross-correlation of `script/eval_real_gt.py` ($r = 0.93$ to 0.99 on `realv7`, 0.61 on the sparse `realv6` export), scoring is that script's planar procedure at the tracker's own samples, and the one off-path `realv6` sample is removed by the rule the realv6 analysis fixed.

The rig's wheelbase, 0.46 m, is recorded and unused: with no steering measurement there is nothing to put through a bicycle model, so the yaw-rate observation that gives the simulation EKF its bias state does not exist, and that filter would reduce to this integration.

Datasets. The four tracker-truthed real recordings of Experiment 3's real-robot section: routes of 22–37 m over 97–219 s. Sources: `output/compare/real/exp1-wheel/` (`wheel_metrics.csv`, `manifest.json`), estimates under `output/output_wheel/real/`.

Results. Table 1.4 and figure 1.2. On the fit recording the baseline reads 0.06 m ATE with a path ratio of 0.99, which is the calibration. Held, it reads 0.26 and 0.22 m on `realv7_001` and `_002` with path ratios of 0.99 and 1.01, and 0.46 m on `realv6_000` against its sparse export, with no jump and a relative error at 1 s of 0.01–0.02 m.

Table 1.4: Wheel+gyro dead reckoning on the real recordings against the optical tracker: the feedback speed scaled by k (fitted on `realv7_000` by minimising planar ATE, held on the others), heading from the integrated gyro with its rest bias removed, rigid $SE(2)$ alignment with scale fixed, lever arm zero and unmeasured. ATE at the tracker's samples; path ratio is estimated over true path length.

Recording	Role	k	Offset [s] (r)	Samples	ATE RMSE [m]	ATE max [m]	Path ratio	$\text{RPE}_{1\,\text{s}}$ [m]
`realv7_000`	fit	0.325	-2.57 (0.99)	491	0.057	0.11	0.990	0.008
`realv7_001`	held	0.325	-3.77 (0.98)	632	0.259	0.47	0.988	0.021
`realv7_002`	held	0.325	-2.60 (0.93)	499	0.219	0.27	1.006	0.008
`realv6_000`	held	0.325	-4.14 (0.61)	233	0.463	0.70	1.041	0.015

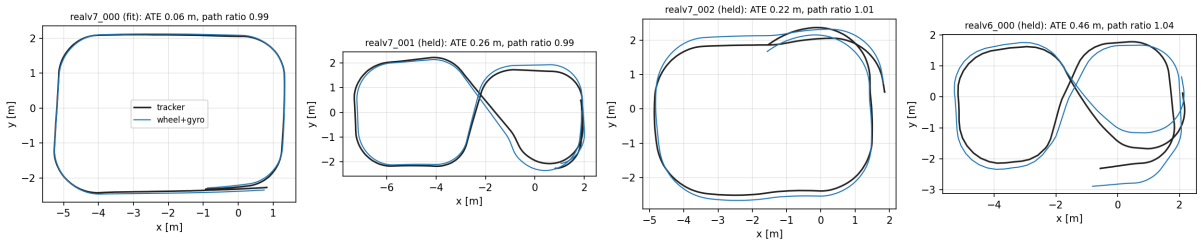


Figure 1.2: Wheel+gyro dead reckoning (blue) against the tracker (black) after rigid planar alignment, one panel per recording; the first is the recording the speed scale was fitted on.

Analysis. The comparison this slot exists for now has a number on both sides, and it does not favour vision. On the same recordings the best ORB-SLAM3 configuration reads 0.44 m on `realv7_001`, 0.47 m on `realv7_002` and 0.46 m on `realv6_000` (Experiment 3, real-robot section, after the scorer's frame repair); the scaled dead reckoning reads 0.26, 0.22 and 0.46 m, with path ratios within 4 % of one where the visual runs read 5–15 % long.

Three things bound that reading. The scale is a fitted constant, not a measured one, so the baseline has had one parameter tuned against the truth it is scored on, on a different recording; a second calibration route, the wheel circumference and the gear ratio, would remove that. The routes are 22–37 m with the robot at 0.2 m s^{-1}: a gyro drifting at the measured rest bias of 0.003–0.006 rad s^{-1} accumulates 0.3–1.3 rad over a run if the bias is not removed and under 0.1 rad after the rest-window removal, and on a longer route that term grows while the visual error need not. And the lever arm is zero for both sides. Against the project's targets no run approaches 0.05 m except the calibration itself; every run is inside the 1.5 m RPE bound.

[Open: the realv6 scale] On `realv6_000` the path-length ratio gives $k = 0.43$ where the three `realv7` bags give 0.32 to 0.33, yet the held $k = 0.325$ scores a path ratio of 1.04 on the matched samples. The export is at 1.7 Hz and fails the validity rules, and the recording predates `realv7` by three days; whether the firmware unit changed or the sparse export misstates the path length is not decided, and the `realv6` row is a diagnostic.

Summary. With one scale fitted on one recording, wheel+gyro dead reckoning localises the real robot to 0.22–0.26 m on the two held `realv7` recordings and 0.46 m on `realv6`, at least as well as the best visual run on each. On routes of this length vision has not earned its place over the proprioceptive sensors; the simulation slot said the same of the measured-noise filter.

Next action. Replace the fitted scale with a measured one (wheel circumference and gear ratio, or a tape-measured straight run); record the steering angle as a measurement so that the bicycle model and the EKF's bias state can be formed with the 0.46 m wheelbase; measure the marker-to-IMU lever arm; and record a route long enough for gyro drift to show.